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Counterevidence of crime-reduction effects from federal grants of military equipment to local police

Anna Gunderson[1], Elisha Cohen[2], Kaylyn Jackson Schiff[2], Tom S. Clark[2]✉, Adam N. Glynn[2,3] and Michael Leo Owens[2]

In 2017, the Trump Administration restored local law enforcement agencies' access to military weapons and some other types of surplus military equipment (SME) that had been prohibited by the Obama Administration. The Justice Department background paper used to justify this decision cited two papers published by the American Economic Association. These papers used SME data collected with a 2014 Freedom of Information Act request and concluded that SME, supplied to local law enforcement by the federal government via the 1033 Program, reduces crime. Here we show that the findings of these studies are not credible due to problems with the data. Using more detailed audit data on 1033 SME, we show that the 2014 data are flawed and that the more recent data provide no evidence that 1033 SME reduces crime.

Militarization is a prominent development in policing across the globe[1–6]. In the United States, it has followed a reorientation of policing towards 'police-invoked law enforcement', whereby "the police themselves initiate the action in the specific instance, though sometimes in response to a general public concern over the problem, and whatever action they take is on their own authority."[7,8] Political and public debate about militarization often focuses on a number of choices and actions by police departments and sheriffs' offices that increase their possession and deployment of military-grade or military-style equipment and attire[5,9–11]. Of course, the increased possession, display and use of military or military-style equipment by subnational law enforcement agencies (LEAs) has been underway for decades. Moreover, there is a history in the United States of occasionally using military forces for domestic policing at local level (for example, colonial militias, US Army control of the South during Reconstruction, National Guard deployments during the Civil Rights Movement, protests against the Vietnam War, and urban and rural unrest and riots)[4–6,12]. However, current trends have been particularly facilitated by federal programs that are intended to support local LEAs. Among the most important federal policies and programs that have contributed to police militarization is the provision of SME to local police and sheriffs via the 1033 Program. That program is administered by the Law Enforcement Support Office (LESO) within the Department of Defense and has its origins in the War on Drugs. However, it has evolved over time to allow local agencies to request SME for virtually any law enforcement function, including both tactical and non-tactical supplies[4–6,11,13–16]. Today, at least 7,000 LEAs participate, according to the LESO, and as of 2017 approximately US$7 billion worth of SME had been transferred to subnational LEAs[17]. This equipment ranges from 'high-visibility' and tactical goods immediately associated with military purposes to more general goods. For example, agencies can access vehicles and aircraft, but they can also access medical supplies or gym equipment. A majority of SME transferred comprises those more general supplies.

Despite the varied nature of the SME items the 1033 Program transfers, and that most SME items obtained by local law enforcement are not tactical, many policymakers justify the 1033 Program on the claim that transfers of SME improve the quality and efficacy of local law enforcement activities. In particular, the Trump Administration used that claim in 2017 to reverse the 2014 restrictions the Obama Administration placed on the program, which banned transfers of military-grade weapons and some other prominent military equipment. Given the policy claim and its influence, and the 1033 Program's importance in contributing to militarization of local police and sheriffs, social scientists seek to identify the positive (and negative) effects of the program.

Generally, however, rigorous empirical studies of the causes and consequences of the variation in the transfers of SME to local LEAs via the 1033 Program are nascent[18–27]. This is unfortunate, because whether the militarization of local LEAs, particularly through the 1033 Program, especially affects crime in cities and counties is an important empirical matter, one with civic, policy and normative implications. One reason for the lack of evidence about the effects of the 1033 Program on US policing is the lack of public, high-quality data on SME transfers. Indeed, it was only in 2014 that the federal government made public 1033 Program data, though the data the government released were crudely aggregated. We show that those data were fundamentally flawed for empirical research and public policymaking. Since the release of the 2014 data, the two most prominent and cited studies, published together in 2017, report a range of positive effects from the 1033 Program[22,23]. In both studies, which received broad media and policymaker attention, the research question is the effect of the aggregate amount of SME—value and quantity of items—sent to a county. The reported benefits include reductions in all types of crime, without negative effects on case closure rates, arrest rates or assaults and injuries to officers, as well as reductions in citizen complaints against the police and declines in civilian assaults and deaths of police officers[22,23].

More granular agency-level data on SME from the 1033 Program exist to allow scholars to better assess the robustness of past empirical findings. It has been publicly available for a while, though few scholars have accessed it. Others may have been unaware of its existence. A notable exception that relies on these less crudely aggregated agency data reports that the 1033 Program has an even larger effect on crime reduction than what earlier studies found[20]. In terms

[1]Department of Political Science, Louisiana State University, Baton Rouge, LA, USA. [2]Department of Political Science, Emory University, Atlanta, GA, USA. [3]Department of Quantitative Theory and Methods, Emory University, Atlanta, GA, USA. ✉e-mail: tom.clark@emory.edu

of research design, that study, which is unpublished, is closest to our study here, which similarly uses sub-county measures of SME transfers to assess the effects of the 1033 Program on crime rates. However, that study uses a different approach to estimate the causal effect of SME than the published studies we revisit in our analysis. Moreover, that study argues that a substantial portion of the estimated effect is due to displacement of crime to other localities rather than absolute reduction in crime. Additionally, other studies using agency-level data, both published and unpublished, and a variety of approaches report evidence of a relationship between SME from the 1033 Program and civilian deaths[28,29]. To the extent the evidence suggests that SME leads to a displacement of crime, along with an increase in violence against civilians, the collective results of these agency-level studies present a potentially troubling picture of the effects of SME on public safety.

Indeed, scholars motivated by this type of concern have gone beyond the proximate effects of SME to report effects as theoretically tenuous as a decrease in suicide rates[30] or reductions in charitable giving and volunteering among Black Americans[31]. Other studies observe that police militarization may produce adverse effects for communities[20,24–26]. Most notably, an analysis of the deployment of special weapons and tactics (SWAT) teams finds that police militarization neither reduces crime nor decreases harm to officers by civilians but may harm public attitudes about the police (for example, support for funding or perceived legitimacy, confidence and trust in the police)[26].

There are at least three explanations for the conflicting results of previous studies of the effects of the militarization of local law enforcement. First, the form of militarization differs across studies. This is most obvious when comparing studies focusing narrowly on SWAT data versus studies that rely on 2014 SME data from the 1033 Program[22,23,26]. However, even among the studies that solely use 1033 Program data, there can be differences in findings due to the existence of multiple categories of SME, including weapons, optics, radar and vehicles. Local LEAs' stock of 1033 Program equipment varies greatly across these categories, with some categories of equipment yielding different effects on, for instance, crime rates. Both of the studies we focus on below disaggregate total SME by key categories and report differential effects by categories of equipment.

Second, empirical studies have used different units of analysis. As noted, the most prominent studies analyse county-level relationships between SME and public safety outcomes[22,23]. However, because the majority of decisions regarding the procurement of SME are made below the county level, at the level of independent local LEAs, the use of county-level data raises a concern of ecological bias. In particular, if crime rates are declining for urban agencies and SME transfers are large for suburban agencies in the same counties, then county-level analysis would mistakenly attribute the urban crime reduction to the suburban use of SME. Hence, county-level measures of SME create a serious challenge to inference, a key issue we discuss in detail below. It is in this spirit that the more recent studies making use of agency-level data advance the study of SME[20,24,29]. However, at the same time, other studies aggregate even further to state level[21,25,32,33]. For instance, one study makes the case for a state-level analysis based on states' oversight, accountability and use of SME from the 1033 Program[31]. However, it is unclear how much of a role or influence states have in the decision by local LEAs—city and county police departments and sheriffs' offices—to apply for 1033 Program aid[34].

Third, while most studies of local law enforcement militarization focus on the distribution of SME from the 1033 Program (for the exception, see Mummolo[26]), they tend to measure militarization differently. Studies can use the quantity of SME the 1033 Program transfers to agencies, municipalities and counties, or they can use the monetary value of SME transferred to localities[22,23]. In addition to these three reasons that we identify for inconsistent findings from existing empirical research, a fourth reason may best explain the disparate findings: the 2014 federal data from the 1033 Program, based on a Freedom of Information Act (FOIA) request by National Public Radio (NPR), are deeply flawed.

However, the federal government does collect cleaner, more comprehensive, and disaggregated data. Given the availability of these richer data, we have two related and original goals in this article. First, we assess the validity of government records about SME transfers to local LEAs. Second, we test the robustness of the high-profile empirical findings and revisit a main empirical claim on which federal policymaking has rested, and even altered federal policymaking across presidential administrations—that SME transfers from the 1033 Program reduce crime. Our findings suggest that any extant conclusions that the transfer of SME via the 1033 Program to cities and counties causes lower crime rates are likely driven by faulty data. We also demonstrate key barriers to generating credible assessments of the effects, whether positive or negative, of SME support for local law enforcement.

We focus exclusively on the effects of SME on crime rates for a number of reasons. First, it is that effect that has featured most prominently in political and policy debates. As noted above, the claim that the 1033 Program reduces crime was at the centre of the Trump Administration's justification for undoing Obama-era limitations to the transfer of SME to local police and sheriffs. And, it is that finding that scholars have most consistently analysed across different empirical studies[20,22,23]. Second, many of the civic outcomes that previous studies have considered at the county or state level are either difficult or impossible to measure at local level. As it is, accurately measuring crime rates is a particularly fraught undertaking[35]. Indeed, the citizen complaint data reported in one previous study are, in fact, not even aggregated at the county level but rather selected from a non-random single jurisdiction from within each county[23]. (We learned this through personal correspondence with the authors of the original study.) Therefore, to the extent that our analysis calls into question prior empirical findings about crime rates, we believe there is room for future work beyond the scope of a single study to revisit the entire literature on the effects of SME on public safety and other civic consequences.

Our results contravene extant findings that militarization of police and sheriffs necessarily makes the community better. In particular, in this article, we find no evidence that federal distributions of SME to local LEAs across the United States reduce crime rates, neither violent nor nonviolent crime rates, in the jurisdictions that receive it. Consequently, we posit that, because of serious data problems and debatable methodological choices in prior studies, the empirical foundations on which social scientists, along with policymakers and the public, stand when making causal claims about the effects of the transfers of SME may be no firmer than quicksand.

Results

Our analysis comprises two related tasks. First, we use two different data releases by LESO to assess the quality of government SME records. Second, we assess the robustness of empirical claims about the effects of SME on crime reduction given the data problems we document. As we detail below, there are considerable discrepancies in the data about which LEAs have and use SME. Moreover, while the more recent audit data represent an important improvement over previous records—indicating the agencies that received the equipment, not just the county to which it was shipped—they still suffer from one important limitation, namely that any equipment that was obtained by a department but subsequently destroyed, lost or returned will not appear in these audits, because they only capture what is currently in local law enforcement possession. And, unfortunately, it is not possible to completely reconstruct the data. We attempted to, by filing a request pursuant to the FOIA (case number DLA-DSERV2018-005182; details and response included

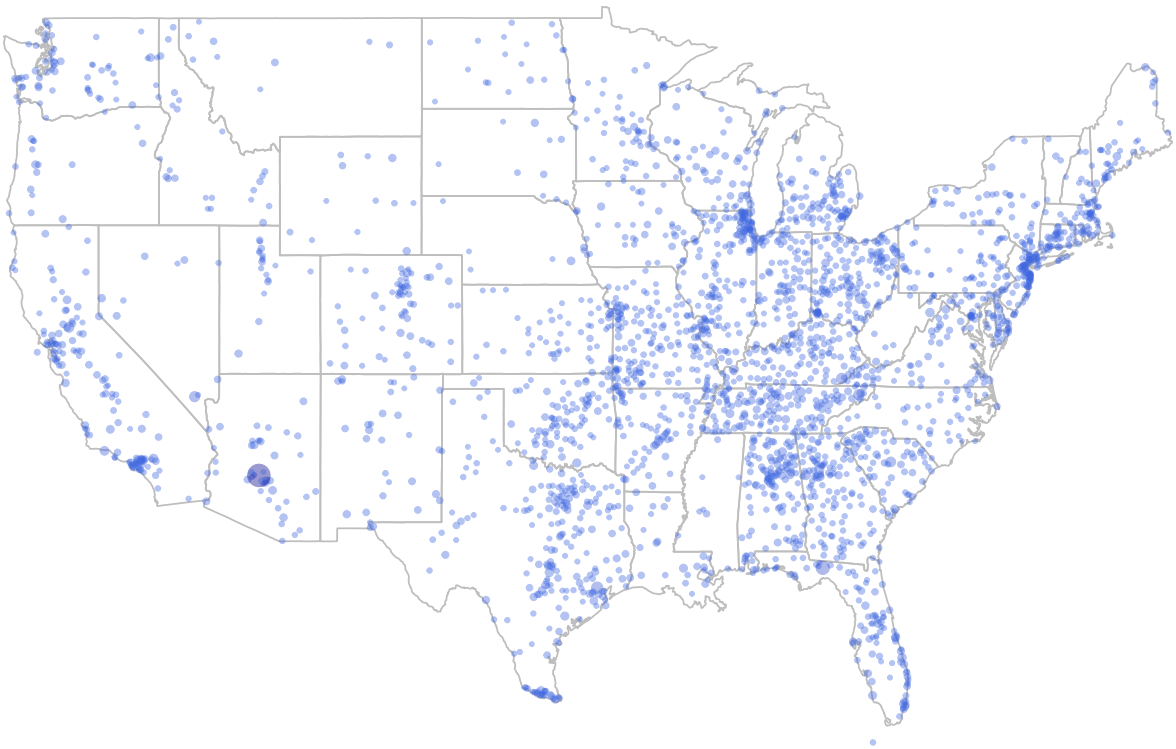

Fig. 1 | Locations of jurisdictions with military transfer aid, 2010–2015. Points show the locations of the 2,855 unique cities to which SME was transferred. Darker colours indicate more military aid transferred. Locations of US states from the US Census Bureau.

in the Supplementary Figures), and requested the agencies that held the accountable items prior to Fiscal Year 2017. In response, the Defense Logistics Agency (DLA) informed us that they did not even record the name of the agency to which the equipment was shipped before Fiscal Year 2017. The DLA does retain records of equipment returned or reported destroyed. However, those records similarly are insufficient to identify who had what at what point in time. For example, from FY14 through FY16, as the federal government began to audit the program more carefully, dozens of LEAs were suspended from participation in the 1033 Program because they had lost or could not account for controlled items which were supposed to be in their possession[34].

Data discrepancies. With the agency-level data in hand, we can better describe the geographic distribution of SME. We are also able to identify a number of problems with the county-level data. That is, we can identify systematic inconsistencies between the two different versions of the data on which scholars have relied.

Figure 1 shows the location of each agency that received a transfer of SME from the DLA between 2010 and 2015. The points show the location of the agency, and the colour corresponds to the total amount, in US dollars, of the value of the transfers to each agency—darker colours indicate more aid. In general, there seem to be some patterns, such as more transfers going to more populous parts of the country. At the same time, there is some interesting variation not necessarily associated with population. The State of Mississippi, for example, stands out for its relative lack of transfers, as do Virginia, North Carolina and West Virginia. There have been a large amount of transfers to Appalachia and the coastal parts of the Mid-Atlantic region.

There could be a number of reasons explaining the geographic variation in the distribution of this equipment. Some past work assumes that the acquisition of equipment is affected by an agency's proximity to one of the major distribution centres, because agencies acquiring SME are responsible for transporting the SME or paying for shipping. In addition, there is variation in the costs to an agency for participating in the program—some states charge a flat fee to individual agencies for administering the program, whereas other states charge a rate based on the assessed value of the SME acquired. Of course, there are also budgetary explanations, such as the agencies' fiscal stress—departments with fewer resources might be compelled to rely on SME more than are departments that have the ability to purchase their own equipment. Related, given the scarcity of high-value items available from the DLA, there is a large random component to whether an agency acquires a piece of equipment, which is affected by the amount of time and energy the agency can devote to searching and writing justifications for equipment. Finally, and especially in the aftermath of the 2016 events we describe above, one might expect that local political conditions influence a department's capacity to make use of the program.

In examining the agency-level data, we uncovered a variety of inconsistencies in the county-level data, those from a FOIA request by NPR in 2014. Table 1 summarizes documented examples of these concerns, which we also describe here in greater detail. Essentially, each of the problems we uncover relates to differences between records of what was shipped and recent inventories. Those discrepancies undermine the positive findings and implications of past studies of the 1033 Program that employed the earlier data.

Missing controlled items. First, in line with high-profile controversies in the media, there are many instances of controlled items for which receiving agencies cannot account. In the county-level data, these items are shown to have been shipped, but the agency to which they were shipped was not recorded. However, the recent inventories also cannot account for the location of these items. Indeed, as mentioned above, in some localities losing track of controlled equipment has become such a problem that LEAs have been suspended from participation in the 1033 Program. For example, Maricopa

Table 1 | Documented problems causing differences between recent LESO inventory data and the 2014 NPR FOIA data

Problem	Description	Example(s)
Missing controlled items	Some controlled items (such as weapons) that were in the NPR FOIA data are missing from more recent LESO inventory data.	In the NPR FOIA data, Baldwin County, AL received four pod-loc bipod support pieces for sniper rifles in 2010. Those items, which are classified in the weapons category, are not included in recent LESO inventory data.
Missing uncontrolled items	Uncontrolled items (tools, office supplies and other items that are not weapons) are removed from the LESO inventory after one year. Therefore, there are many uncontrolled items that were in the NPR FOIA data that are no longer in the LESO inventory.	In the NPR FOIA data, Clark County, IN received 910 feet of radiofrequency cable, a pair of binoculars and a wrench in 2010. Those items are not included in recent LESO inventory data.
Missing counties	Some counties (or county-years) are missing from the NPR FOIA data.	Sanpete County, UT is missing from the NPR FOIA data for 2009 and 2010, but that same county is recorded as having received reflex sights and rifles in 2009 and 2010 in the LESO inventory data. Howard County, MD is also missing from the NPR FOIA data for 2010, but that same county is recorded as having received items in 2010 based on recent LESO inventory data.
Inconsistent item values	Some items identified as matches (having the same description, national stock numbers (NSN) and shipping date) have different values.	In the NPR FOIA data, Yuba County, CA received a truck (NSN 2320-01-107-7153) worth US$47,069 in 2012. In recent LESO inventory data, Marysville Police Department in Yuba County (the only agency in that county to receive a vehicle in that year) received a truck with the same NSN worth US$63,894 in 2012.
Inconsistent item records	Some items identified as matches (having the same description, are in the same county and are the only items of that type shipped to that county in that year) have different NSN and shipping dates, as well as values.	In the NPR FOIA data, Humboldt County, CA received a utility truck valued at US$47,609 with NSN 2320-01-107-7153 that was shipped on 10 December 2012. In recent LESO inventory data, Ferndale Police Department in Humboldt County (the only agency in that county to receive a vehicle in that year) received a utility truck valued at US$89,900 with NSN 2320-01-346-9317 that was shipped on 3 January 2012.
State and federal agencies	It is unclear how items received by state and federal LEAs were recorded in the NPR FOIA data. The inclusion (or not) of state and federal agencies in county totals could be a source of some discrepancies between the NPR FOIA data and more recent LESO inventory data.	In the NPR FOIA data, there is no record of items received by Fulton County, AR in 2011. However, in recent LESO inventory data, the Arkansas State Parks Department, which is geolocated to Fulton County, received 69 rifles in 2011. The Arkansas State Parks Department is the only LEA in Fulton County, AR in the 1033 Program. The 69 weapons do not appear to be elsewhere in the NPR FOIA data.

County Sheriff Joe Arpaio admitted in 2014 that the county could not account for nine firearms it received through the program. At that point in time, a large number of separate agencies were suspended from the 1033 Program[34,36]. It is difficult to determine the extent of this problem in our data, as even discrepancies between the two datasets may be attributable to item returns or destruction rather than true missingness. However, we attempt to show the overall discrepancies in Fig. 2, a scatterplot of the data grouped by national stock number (NSN), county, state and shipping date; That is, each observation is a discrepancy in value for the same item, for each county-day. Only 2% of the data are exact matches (see inconsistent item values below for more information); nearly 78% of the data is on the *y* axis, with a positive value listed in the NPR dataset but no item listed in the LESO dataset, while 7.4% is on the *x* axis, with a positive value listed in the LESO dataset but no item (or an item with zero value; see below) listed in the NPR dataset. While this scatterplot is only an estimate of our data discrepancies (and includes both controlled and uncontrolled items), it highlights the vast differences between the two datasets.

Missing uncontrolled items. LEAs acquire ownership of uncontrolled items after one year and do not need to report possession for inventories. The more recent agency-level data therefore cannot account for all uncontrolled items. As above, it is difficult to evaluate the extent of this problem, but Fig. 2 shows how widespread the data discrepancies are. While missing uncontrolled items is a less theoretically interesting problem, it is an issue of consistency across SME transfer datasets and represents a limitation for working with recent inventory data.

Missing counties. In the county-level data, there are some counties that report no military surplus transfers, even while recent records indicate that they received equipment during the period covered by those data. Possible reasons include the indirect transfer process described above as well as simple errors in the original data released. In total, there are 76 unique counties missing in the NPR FOIA data that we have data for in our dataset, for a total of 1,274 observations, approximately 1.5% of our total dataset. These counties tend to be fairly small (10,000 to 20,000 residents), so it is possible their items are shipped to other jurisdictions with larger populations, attributing those items to a different county in the NPR dataset (Table 2).

Inconsistent item values. There are discrepancies between the value associated with individual items in the original, county-level data and the value reported in the more recent, agency-level data. Discrepancies go both ways—sometimes items were assigned higher values in the original data, while sometimes they are assigned higher values in the more recent data. The consequences of these discrepancies are direct, as a common operationalization of military transfers is the total value (per capita) of aid sent to each locality. We do know the source of one piece of inaccurate information: of the 243,328 observations in the NPR dataset, 11,436 (approximately 5%) have a listed item value of US$0 (items such as motorcycles, night-vision sights and buses). Only 21 observations in the 76,598 observations in our dataset have a listed value of US$0 (mainly binoculars, but also an armoured truck and a new terrain vehicle). In addition to inconsistencies across data releases, we also find that, within versions of the data, identical items are sometimes assigned different values. We are not sure of the

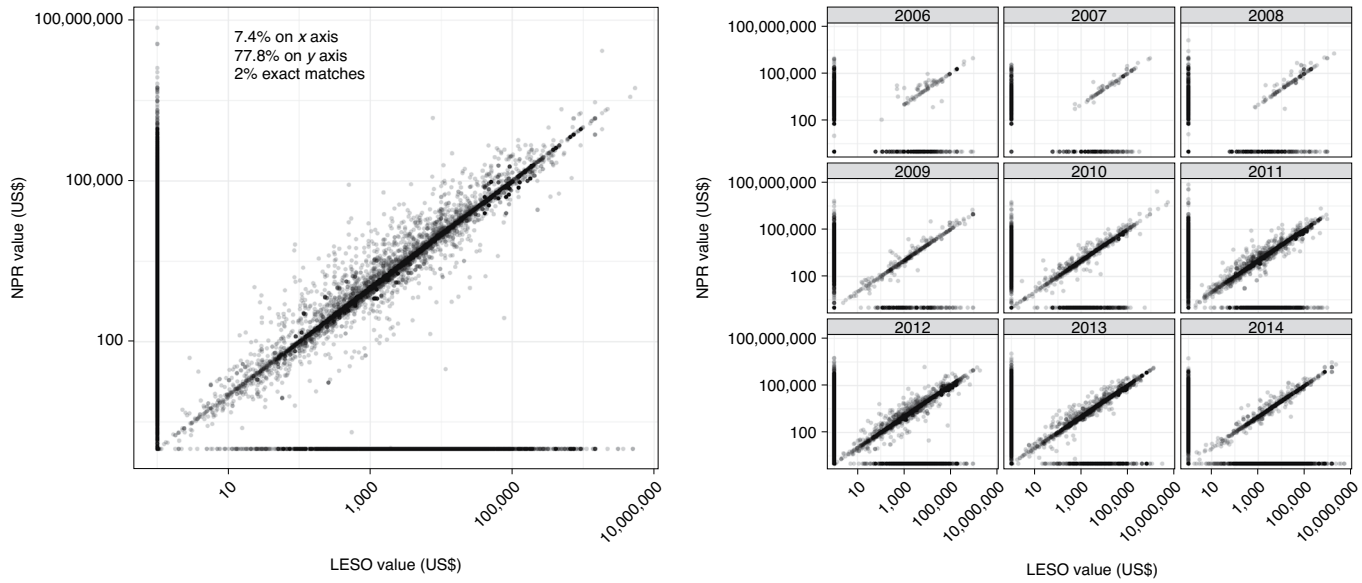

Fig. 2 | Item discrepancies between the NPR FOIA data and the LESO inventory data. The black dots represent a total of 138,283 observations matched on county, state, NSN and shipping date, showing all results from 2006 to 2014 (left) and individual scatterplots by year (right). The exact matches are those items that have a positive value in both datasets (that is, items with a listed value of zero are excluded).

Table 2 | Hypothesized problems causing differences between recent LESO inventory data and the 2014 NPR FOIA data

Problem	Description	Example(s)
Damaged items	Damaged items that are reported to LESO state coordinators are removed from the LESO inventory after some time. The removal of damaged items could cause discrepancies between the NPR FOIA data and recent LESO inventory data.	We have no way of knowing which items are no longer in the LESO inventory because they were damaged.
Transferred items	Items can be transferred legally (approved by LESO state coordinators) or transferred illegally (no approval) between LEAs. These transfers could cause discrepancies between the NPR FOIA data and recent LESO inventory data.	We have no way of knowing which items in the LESO inventory were transferred.
Inconsistent shipping locations	LEAs arrange for the shipping of items obtained through the 1033 Program and have flexibility regarding where the items are sent. The NPR FOIA data are based on shipments to counties, but the more recent LESO inventory data are based on the agencies responsible for the items. Items shipped to a county different from the county of the LEA could cause differences between the two datasets.	Without analysing the individual requests submitted to LESO state coordinators, we cannot identify which (if any) LEAs and counties are subject to this problem.

reasons for these discrepancies, though the degree of this variation can be substantial.

Inconsistent item records. In some instances, the original data report items that have been transferred to a county, but we cannot locate such an item in the more recent inventory, even while we can locate a similar item in that county. In Table 1, for example, we describe two vehicles, one which was reported in the county-level data and the other in the agency-level data, with similar shipping dates, but different serial numbers. Because there are several discrepancies across the two datasets, it is not possible to determine the extent of this problem.

State agencies. Equipment shipped to state agencies, which can ultimately be deployed anywhere in a state, are associated with only the county to which it was shipped. This will result in an inflated level of military aid to counties with state agency receiving locations. We identified non-local agencies by filtering agency names with words such as 'state agencies', 'state departments' (parks and recreations or correctional departments), or 'multi-county or multi-agency task forces'. In our data, there are 125 unique agencies that match these characteristics, for a total of 8,175 observations, accounting for roughly 5% of our dataset.

While we cannot determine the exact cause of these discrepancies, recent media reports and a government study highlight the lack of rigorous controls to track this equipment which can result in (a) unreported transfers of equipment from one agency to another, (b) equipment being lost or damaged but not reported or (c) poor record keeping of the equipment transferred in the first instance. For example, in a sting operation, the Government Accountability Office was able to create a fictitious LEA and gained permission to receive over US$1 million in military surplus equipment[17]. In addition, because the original data only report the county to which equipment is shipped, it is possible that there are items that have been shipped to one county on behalf of an agency located in another county. For example, we have learned of instances in which aircraft and vehicles have first been sent to one location for repair, other maintenance or even training, before being ultimately directed to the agency that has custody of the equipment.

The implication of these various discrepancies is that the data released by the LESO program are unreliable and ill-suited to systematic analysis of the effects of SME on law enforcement outcomes. Most alarming is that only 2% of the data can be matched exactly across the two data releases. In conjunction with the myriad other problems we identify, we have no reason to believe that either version of the data is comprehensive or valid, though there do seem to be greater problems with the county-level data.

Replication analyses. Our replications focus on two prominent studies that were published simultaneously in the *American Economic Journal*. We focus on them because of (a) the broad media coverage of their findings and (b) the use of their findings by the Trump Administration to justify revoking limitations that the Obama Administration imposed on the 1033 Program, including prohibitions against the transfer of certain types of SME to local LEAs. For instance, on 28 August 2017, then-US Attorney General Jeff Sessions spoke before the National Fraternal Order of Police, remarking in defense of the Trump Administration's recent full restoration of the 1033 Program that "Studies have shown [surplus military] equipment reduces crime rates, reduces the number of assaults against police officers, and reduces the number of complaints against police officers. Those restrictions went too far. We will not put superficial concerns above public safety."

Specifically, we conduct the same analyses presented in Bove and Gavrilova (BG)[22] and Harris, Park, Bruce and Murray (HPBM)[23], using the agency-level data. We conduct three separate analyses for each study: first, using the authors' code and data, we replicate their published results exactly. Second, we attempt to replicate their findings using our agency data, aggregated to the county (more details on this in the Supplementary Methods). Finally, we use the more recently released DLA data to estimate the relationship between 1033 Program disbursements and crime rates at the agency level. Both BG and HPBM use instrumental variables in their analysis, and we calculate identical instrumental variables in our data using their methods. The first-stage results for both analyses, as well as the full replication tables, are given in the Supplementary Results. For the original BG analysis, our county-level replication and our agency-level replication, all first-stage results are positive and significant. We also note that both our aggregated county-level and agency-level replications provide larger first-stage estimates, and as in the original BG analyses have F statistics far above 10 (the typical rule of thumb for a sufficiently strong instrument). For the original HPBM analysis, only some of the coefficients on the individual instruments were significant in the first stage of both the item and value analyses, but the F statistic was significant for both analyses. That said, for the original HPBM analysis, the F statistic for the value analysis was below 10. For our aggregated county-level first-stage replication analyses, none of the individual instruments were significant, but as with HPBM, the F statistics were significant for both the item and value analyses. Additionally, the F statistic from our aggregated county-level value analysis was below 10 but larger than that from HPBM, while the F statistic from our aggregated county-level item analysis was just below 10. Finally, for our agency-level first-stage replication analyses, again none of the individual instruments were significant, but the F statistics were significant and well above 10 for both the item and value analyses.

BG seek to understand the effect of militarization aid on a variety of outcomes such as arrest rates, offenders killed and civilian complaints, among others. For our purposes, we focus on their analysis of the effect of military aid on a variety of county crime rates—total, homicide, robbery, assault, burglary, larceny and vehicle theft. To ensure we have complete coverage on the number of jurisdictions that could request equipment from the federal government, in the replication we include all cities and all counties in all states that reported crime data, the main outcome variable of interest. Though there are over 19,000 local jurisdictions across the 50 states, just over 9,000 of those reported crime data to the national government. (Formally, there are 19,354 'incorporated places' in the United States as of 2018.) Therefore, our sample includes the approximately 9,000 agencies reporting crime data which could potentially request SME through the 1033 Program.

Figure 3 summarizes the results of our three analyses: the replicated results from BG, our replicated analysis aggregating the agency-level data to the county level (the same unit as BG) and our analysis run at the agency level. In each analysis, we employ their exact research design, but use different data and/or units of analysis. Our results, both in the replication county estimations and the agency-level estimations, differ substantially from those of BG. We find no clear evidence of a significant effect of military aid on a variety of crime rates. For our preferred agency-level analysis, many of our point estimates are positive (that is, the opposite direction from BG), and those that are negative are insignificant. For our aggregated county-level analysis, most of the coefficients are negative, but insignificant. The two significant negative coefficients (for robbery and burglary) are barely significant and are accompanied by agency-level estimates that are insignificant and of the wrong sign. Furthermore, this pattern of insignificance does not seem to be simply a matter of low power. In the Supplementary Information, we show that our analysis is generally well powered. Additionally, in all crime categories, at least one of our estimates has tighter confidence bounds and higher power than the original BG analysis. Therefore, even if the NPR FOIA data were valid and the BG research design perfect, it is more likely that the BG findings are due to an underpowered design overstating an effect than that our null findings are due to a lack of power[37]. Additionally, using the equivalence test approach of Gross[38] and Rainey[39], we find that we can rule out substantively large effect sizes, as well as all of the significant effects presented in BG (see the Supplementary Results for details).

As with BG, HPBM study how military transfers affect crime rates of varying types. They also analyse the effect on other outcomes, including citizen complaints, assaults on officers and arrest rates. For the purposes of this replication, however, we focus solely on their analysis regarding crime rates. Also as in BG, HPBM identify this relationship at the county level. We again replicate their study using their design and our more recent data. Figure 4 reports the results of this exercise.

Our analysis yields no support for HPBM's finding of significant and negative relationships between the logged number of items given to counties and robbery, assault and vehicle theft rates. We find no statistically significant effect on those, or any other, categories of crime rates. Indeed, when we aggregate data to the county level, we only find qualitatively similar results in the analysis of homicides, though even those results do not support the claim that SME reduces crime. The county-level results seem to be driven by different patterns in jurisdictions within counties receiving more aid relative to those receiving less (or no) aid. When aggregated, that heterogeneity is masked, and one draws the incorrect conclusion that places receiving more aid experience lower crime rates. Similarly, if we instead examine the logged value of items, HPBM find a significant and negative relationship between military equipment transfers and robbery, assault and vehicle theft rates, whereas we find no significant negative effect of the value of military equipment transfers on a variety of crime rates.

Specifically, in all of these analyses, we typically find no evidence of an effect, and in our preferred agency-level analysis, many of our point estimates are positive (that is, the opposite direction from HPBM), and those that are negative are insignificant (with the exception of the value-based analysis for robbery). For our aggregated county-level analysis, most of the coefficients are negative, but insignificant. The two significant negative coefficients (for

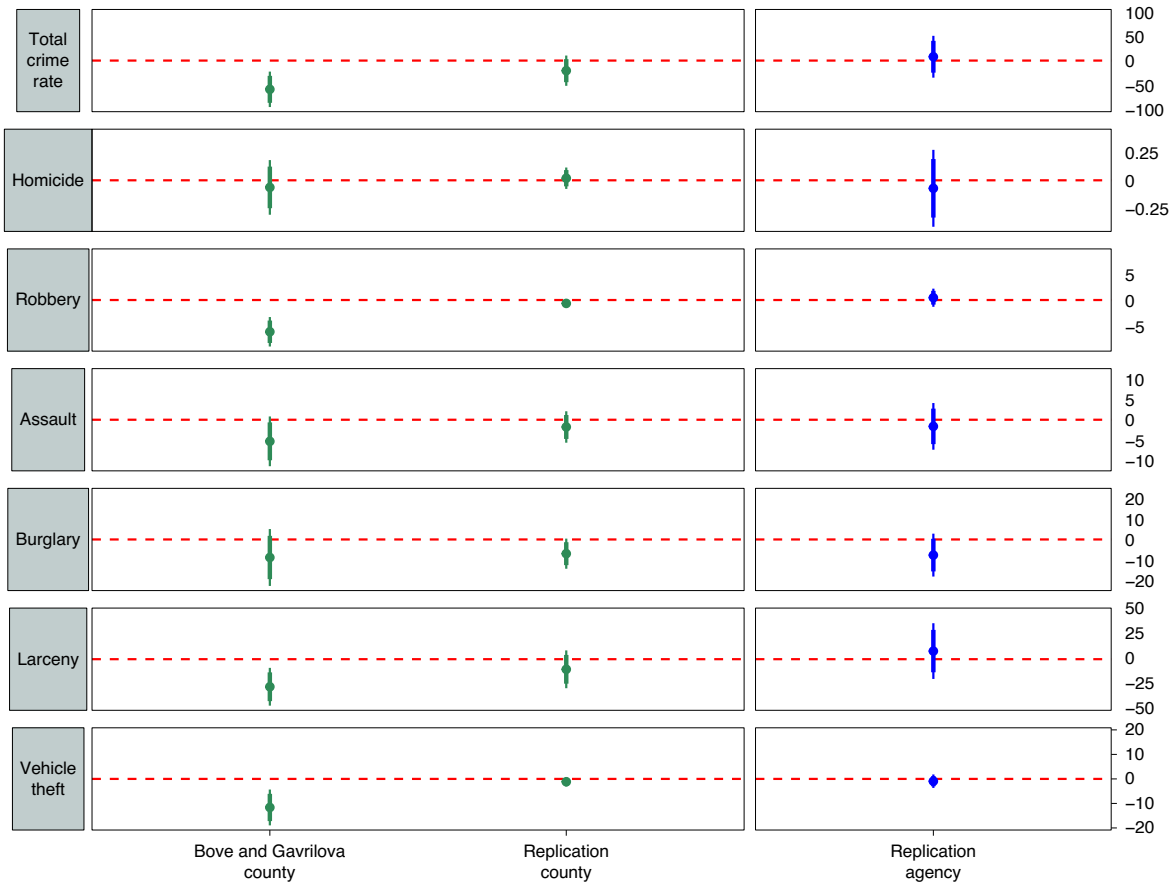

Fig. 3 | Replication of BG analyses using both original and more recent data. County-level (left) and agency-level (right) analyses. Points are point estimates, and bars are 95% (two-tailed) confidence intervals. The BG estimates correspond to their models and can be matched back to the results in their Table 2. The BG results suggest a statistically significant negative impact of lagged total military aid on five out of seven crime outcomes (total crime rate, robbery, assault, larceny and vehicle theft). However, our county-level replication results suggest a statistically significant negative impact of lagged total military aid on only two crime outcomes (robbery and burglary). Our agency-level replication results suggest no statistically significant impact of lagged total military aid on any crime outcomes. For BG, total crime rate: $\beta = -59.29$, $P < 0.001$, 95% CI = [−87.1, −31.48]; homicide: $\beta = -0.06$, $P = 0.506$, 95% CI = [−0.25, 0.12]; robbery: $\beta = -6.1$, $P < 0.001$, 95% CI = [−8.26, −3.95]; assault: $\beta = -5.31$, $P = 0.03$, 95% CI = [−9.96, −0.65]; burglary: $\beta = -8.75$, $P = 0.109$, 95% CI = [−19.26, 1.76]; larceny: $\beta = -27.43$, $P < 0.001$, 95% CI = [−41.73, −13.13]; vehicle theft: $\beta = -11.64$, $P < 0.001$, 95% CI = [−17.2, −6.08]. For our county-level replication, total crime rate: $\beta = -20.81$, $P = 0.092$, 95% CI = [−44.57, 2.94]; homicide: $\beta = 0.02$, $P = 0.617$, 95% CI = [−0.05, 0.09]; robbery: $\beta = -0.65$, $P = 0.005$, 95% CI = [−1.08, −0.22]; assault: $\beta = -1.78$, $P = 0.241$, 95% CI = [−4.72, 1.16]; burglary: $\beta = -6.9$, $P = 0.019$, 95% CI = [−12.46, −1.34]; larceny: $\beta = -10.06$, $P = 0.174$, 95% CI = [−24.37, 4.25]; vehicle theft: $\beta = -1.16$, $P = 0.124$, 95% CI = [−2.61, 0.29]. For our agency-level replication, total crime rate: $\beta = 7.91$, $P = 0.639$, 95% CI = [−24.92, 40.73]; homicide: $\beta = -0.07$, $P = 0.594$, 95% CI = [−0.33, 0.19]; robbery: $\beta = 0.44$, $P = 0.516$, 95% CI = [−0.88, 1.77]; assault: $\beta = -1.63$, $P = 0.468$, 95% CI = [−5.99, 2.73]; burglary: $\beta = -7.6$, $P = 0.066$, 95% CI = [−15.52, 0.33]; larceny: $\beta = 8.06$, $P = 0.457$, 95% CI = [−13.02, 29.14]; vehicle theft: $\beta = -0.93$, $P = 0.387$, 95% CI = [−3.02, 1.16]. The replication regressions are run on county data with 15,683 observations from 2010 to 2014 and agency data with 45,331 observations from 2010 to 2015. The BG county data are from 2006 to 2012 and include 17,822 observations. All regression specifications control for percent in poverty, logged median household income, unemployment, logged population, share male, share Black, share aged 15–19 yrs, share aged 20–24 yrs, share aged 25–34 yrs, and agency/county and year fixed effects. We removed from the agency-level database outliers that had total crime rates of over one million.

robbery and vehicle theft in the value-based analysis) are barely significant. Notably, the agency-level vehicle theft estimate is insignificant. Again, this pattern of insignificance does not seem to be simply a matter of low power. In the Supplementary Information, we show that our analysis is generally well powered. In every case where HPBM had a significant negative finding, our county analysis is insignificant with tighter confidence bounds and more power. Therefore, again even if the NPR FOIA data were valid and the HPBM research design perfect, it is more likely that the HPBM findings are significant due to an underpowered design overstating an effect than that our null findings are due to a lack of power. The value-based robbery findings are the only case where our analyses seem to replicate the HPBM finding. However, we are sceptical of any results relying on the value of SME, because they compound errors in the data on both the number of items shipped and their assigned values. As we show in the Supplementary Information, there is considerable variation in the values of identical items in the various versions of data available. Additionally, using the equivalence test approach of Gross[38] and Rainey[39], we can rule out substantively large effect sizes, as well as all of the significant effects presented in HPBM with the exception of the value-based robbery finding discussed above (see the Supplementary Results for details).

Even if we limit the analysis to only the years in which we have overlapping data between HPBM, BG and our data (from 2010 to 2012),

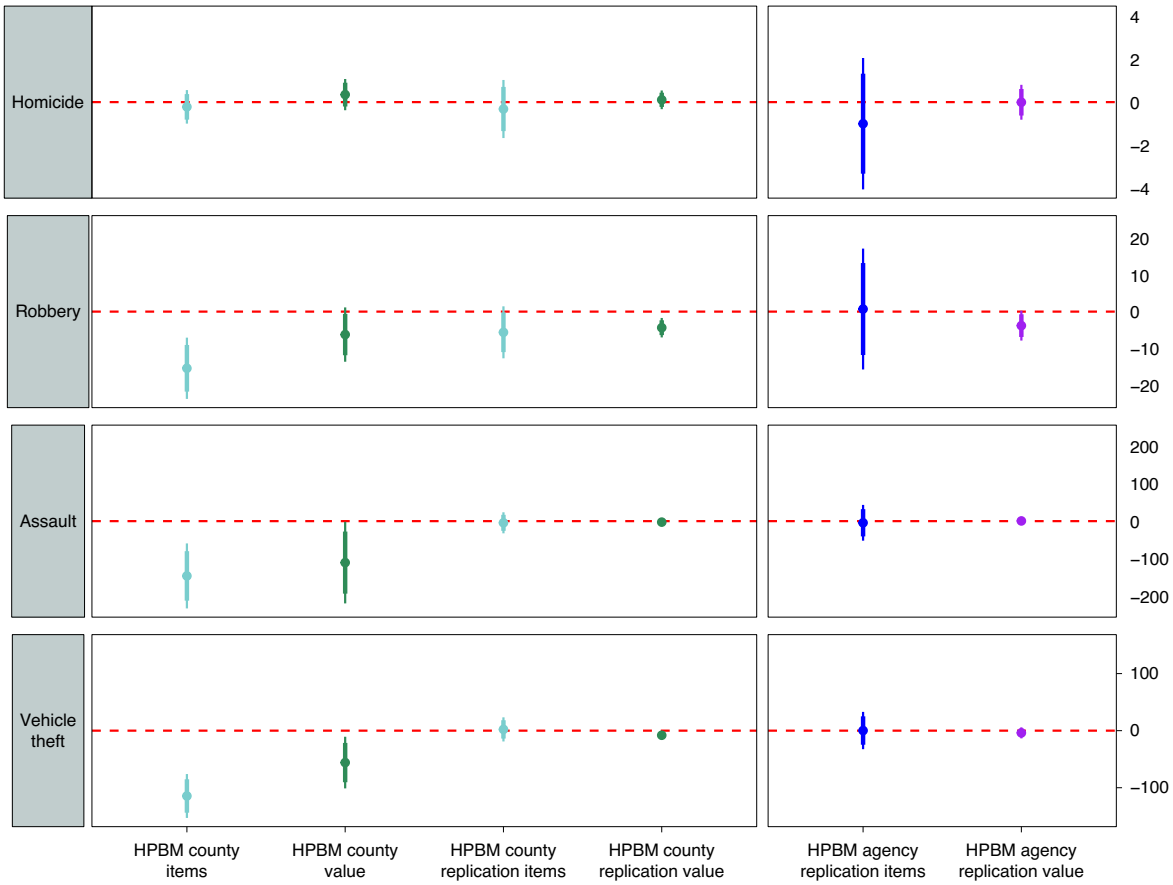

Fig. 4 | Replication of HPBM analyses using both original and more recent data. County-level (left) and agency-level (right) analyses. Points are point estimates, and bars are 95% (two-tailed) confidence intervals. The HPBM estimates correspond to their models and can be matched back to the results in their Table 8. The HPBM results suggest statistically significant negative impacts of lagged military aid items and value on three of four crime outcomes (robbery, assault and vehicle theft). However, our county-level replication results suggest statistically significant negative impacts of lagged military aid items and value only on robbery, and of aid value only on vehicle theft. Our agency-level replication results generally suggest no statistically significant impacts of lagged military aid items and value on any crime outcomes, except for the impact of aid value on robbery. For HPBM and aid items, homicide: $\beta = -0.22$, $P = 0.466$, 95% CI = [−0.81, 0.37]; robbery: $\beta = -15.39$, $P < 0.001$, 95% CI = [−21.73, −9.06]; assault: $\beta = -145.8$, $P < 0.001$, 95% CI = [−211.6, −79.99]; vehicle theft: $\beta = -114.51$, $P < 0.001$, 95% CI = [−143.76, −85.25]. For HPBM and aid value, homicide: $\beta = 0.35$, $P = 0.216$, 95% CI = [−0.2, 0.9]; robbery: $\beta = -6.23$, $P = 0.03$, 95% CI = [−11.85, −0.62]; assault: $\beta = -110.35$, $P = 0.009$, 95% CI = [−192.97, −27.73]; vehicle theft: $\beta = -55.94$, $P = 0.001$, 95% CI = [−90.37, −21.5]. For our county-level replication and aid items, homicide: $\beta = -0.32$, $P = 0.54$, 95% CI = [−1.35, 0.71]; robbery: $\beta = -5.63$, $P = 0.04$, 95% CI = [−11, −0.25]; assault: $\beta = -4.44$, $P = 0.683$, 95% CI = [−25.74, 16.86]; vehicle theft: $\beta = 2.19$, $P = 0.789$, 95% CI = [−13.87, 18.26]. For our county-level replication and aid value, homicide: $\beta = 0.11$, $P = 0.529$, 95% CI = [−0.22, 0.44]; robbery: $\beta = -4.37$, $P < 0.001$, 95% CI = [−6.38, −2.36]; assault: $\beta = -2.63$, $P = 0.419$, 95% CI = [−9, 3.74]; vehicle theft: $\beta = -8.07$, $P = 0.003$, 95% CI = [−13.42, −2.71]. For our agency-level replication and aid items, homicide: $\beta = -1$, $P = 0.397$, 95% CI = [−3.33, 1.32]; robbery: $\beta = 0.7$, $P = 0.912$, 95% CI = [−11.79, 13.2]; assault: $\beta = -4.4$, $P = 0.812$, 95% CI = [−40.63, 31.82]; vehicle theft: $\beta = 0.2$, $P = 0.987$, 95% CI = [−24.67, 25.07]. For our agency-level replication and aid value, homicide: $\beta = -0.01$, $P = 0.982$, 95% CI = [−0.62, 0.61]; robbery: $\beta = -3.81$, $P = 0.016$, 95% CI = [−6.89, −0.72]; assault: $\beta = 0.53$, $P = 0.898$, 95% CI = [−7.55, 8.61]; vehicle theft: $\beta = -3.89$, $P = 0.3$, 95% CI = [−11.24, 3.46]. The replication regressions are run on county data with 15,460 observations from 2010 to 2014 and agency data including 45,055 observations from 2010 to 2015. The HPBM county data are from 2000 to 2013 and include 36,671 observations. All regression specifications control for lagged arrest rates, economic controls, and agency/county and year fixed effects. We removed from the agency-level database outliers that had total crime rates of over one million.

the purported negative relationship between transfers of military equipment and crime at local level does not appear, either in the county-level or agency-level analysis. The HPBM sample size decreases, however, as their original data covers 2000 to 2013, so the results could be partially due to a decrease in statistical power. While it is important to note that our analyses use different time periods, the results here suggest that the conclusions from both BG and HPBM are sensitive to the selection of either the time period or unit of analysis, or both. We therefore cannot conclude definitively that military aid exerts an influence on crime rates, at either the county or agency level.

Discussion

Militarization is a part of modern policing. In the United States, intergovernmental aid, particularly the transfer of SME from the Department of Defense to police departments and sheriffs' offices, contributes to the militarization of local law enforcement. Such equipment includes vehicles, weapons and attire more traditionally associated with military combat and interstate conflict, along with a range of other surplus goods.

Militarization of policing raises normative concerns. Many concerns in the United States involve the consequences of the shift in how police appear and behave, especially when their appearance

and behaviour are associated with intergovernmental transfers of SME. A key concern is whether SME transfers to police and sheriffs are consequential for public safety, either positively or negatively. After all, the primary policy assumption for SME transfers is that they produce a net positive for public safety.

Past empirical studies of police militarization in the United States have concluded that SME transfers to local police and sheriffs cut crime[22,23]. They claim, too, that such transfers reduce injuries to officers and have no adverse effects on civilian injuries or complaints about local law enforcement officer behaviour. Our study, however, which includes replication of key studies and leverages refined data measuring SME transfers, invites reconsideration of prior conclusions and claims.

We highlight the measurement problems of prior data on SME transfers via the 1033 Program, which the DLA maintains and shares. We also produce empirical findings that contradict key extant studies. As a result, our analysis counsels against drawing firm conclusions and promoting claims about the positive policy efficacy of police militarization in general, and via the 1033 Program in particular, especially for crime rates, which is the most important policy concern.

Based on our empirical findings, we recommend that policymakers and social scientists seeking to make reliable claims about police militarization employ more recent and limited but far cleaner and more valid data on SME transfers[26]. Additionally, our findings suggest that scholars and policymakers would benefit from a more rigorous effort to obtain systematic data on the kinds of accountable, controlled, military-grade equipment that local LEAs possess, both SME from the 1033 Program and other equipment purchased via financial subsidies of other government programs (for example, Urban Area Security Initiative grants) and local source revenue. That is important: just as we lack systematic, comprehensive and accessible data on police use of force against civilians, we lack similar data on the kinds of enhanced or high-level military-oriented equipment supplied to police and sheriffs by and beyond the US military.

Improving the empirical basis for justifying (or critiquing) police militarization is important. As past studies of the United States recognize, the civic (and political) effects of policing, generally, are widespread but often more deeply felt by some communities relative to others. Police officers and sheriffs are ubiquitous representatives of the state and, therefore, have an out-sized effect on how citizens perceive and experience government. While we focus here on the effects of police militarization on crime rates, our results should also cause us to question similar analyses that claim to find no evidence of deleterious effects of militarization on citizen perceptions of and interactions with the police.

Methods

We seek to assess the quality and validity of government SME transfer records and to replicate and evaluate the robustness of high-profile empirical findings that claim that SME transfers from the 1033 Program reduce crime. The sections below detail the relevant processes and estimation strategies used for each of these components of our analysis, with further detail provided in the Supplementary Methods. Note that all statements of statistical significance are based on two-tailed tests.

Assessing the quality and validity of SME transfer records. The most prominent studies of the effects of SME rely on data that report items shipped from the DLA and the county in which the recipient was located. This dataset was released by the LESO of the DLA in response to a FOIA request by NPR. However, the government also requires that LEAs report quarterly inventories of the SME they have in their possession, a process that was formalized and required by law by the FY2017 Defense Authorization Act. The availability of these agency-level data allows us to not only compare the county-level records from the NPR FOIA data used by previous studies with the agency-item LESO inventory, but also to generally assess the quality, reliability and validity of the LESO SME transfer records. The DLA did not retain records of agency inventories before 2014; therefore, for earlier years, we must reconstruct the inventories using what information is contained in subsequent inventories. For those years, therefore, our estimates of SME for each agency are a lower bound on how much SME each agency had.

Reliability concerns. Through attempting to match item records between these datasets and through communication with Kenneth (Ken) MacNevin, the Chief Spokesperson for DLA Disposition Services, we identified the many inconsistencies between the NPR FOIA data and the LESO inventory data discussed in this article and detailed in Tables 1 and 2 and Fig. 2 (more information about the complexities in working with this militarization data is provided in the Supplementary Methods). To more thoroughly document the extent to which these inconsistencies arise, we also match on NSN, county, state and shipping date to see how the values for the same items shipped on the same day to the same county differ across datasets. In the data, only 2% are exact matches (having positive values of aid in both datasets), though nearly 8% are exact matches when we include observations in the NPR dataset with an item value of zero (that is, there is an item for that county–NSN–day in the NPR dataset with a listed value of zero, but we do not have any item listed for that day). Nearly 78% of the data is on the y axis (no record in the LESO dataset but a positive record in the NPR dataset), while 7.4% is on the x axis (a positive value in the LESO data but nothing listed in the NPR data). While the records missing from the LESO inventory dataset can be explained by uncontrolled items dropping off of the inventory after a year, as is the procedure for relinquishing control of those generic office equipment items to the LEA, the items missing from the NPR dataset but in the LESO inventory suggest important inconsistencies between the controlled items recorded in the two datasets. Moreover, Supplementary Figs. 4–7 map the differences between the military aid data used by BG and HPBM, revealing inconsistencies, likely differences in interpretation and other potential problems even in working with the (assumed) same county-level SME transfer records. These inconsistencies and the other problems that we document suggest caution in working with these records of SME transfers.

Ecological fallacy concerns. Furthermore, a particular concern in working with the NPR FOIA data is that all item shipment records provide only the name of the county to which the equipment was shipped, rather than indicating the specific local LEA that requested the item. The immediate cause of concern, then, is a matter of ecological fallacy. In this setting, an ecological fallacy would occur if, for example, military aid was going disproportionately to more suburban or rural parts of a county, while crime rates were decreasing in the more urban parts of the county, or vice versa. A county-level analysis would indicate a positive correlation between military equipment and crime reduction, while an agency-level analysis would indicate a negative correlation between military equipment and crime reduction. Note, the concern here is not about ecological displacement, where the effect of militarization simply shifts the location of crime elsewhere. The concern about ecological fallacy can be driven by heterogeneity among jurisdictions in changes in crime rates (even unrelated to militarization), which is masked when agency-level data are aggregated to the county level. Under conditions such as the example we offer, inaccurate inferences about those jurisdictions may follow from analysing county-level data. The agency-level data provided by the LESO inventories avoid this problem and allow us to replicate the results of previous studies at both the county and agency level, described more below. Differences in the county-level results and the agency-level results suggest ecological fallacy concerns, as well as the influence of the data inconsistencies that we describe.

General replication approach. We replicate and extend the two most prominent studies of the effects of the 1033 Program, namely BG and HPBM, using more recent, agency-level SME transfer data. Thus, the sources of data that we use are the authors' datasets and replication code, the agency-item dataset representing LESO's 1033 Program inventory that we downloaded on 31 March 2018, the Federal Bureau of Investigation (FBI)'s Uniform Crime Reporting (UCR) Program data used by the previous studies and demographic data from the Census Bureau also used by the previous studies. With the LESO inventory data, we construct an agency-level dataset and also aggregate to the county level for a county-level dataset (more information about the construction of our datasets is provided in the Supplementary Methods).

Sample and time period. To use the LESO inventory data, we first needed to match the agency-level data to jurisdictions. We used the Google Maps application programming interface to search for the name of each agency and retrieve its address, locality, latitude and longitude. These pieces of information allow us to link the agency to a specific jurisdiction (city or town) and county. Then, while the agency-level data provide us with information for agencies participating in the 1033 Program, they obviously do not provide us with information for agencies not participating in the program. In other words, we only have those agencies for which there were any military transfers. We are left to construct the set of jurisdictions that received no military transfers. To overcome this issue, we rely on the universe of cities that report crime rates to the FBI. This is neither the universe of agencies receiving military transfers nor the universe of agencies in the United States. The sample in our data, then, is the set of agencies reporting crime rates to the FBI. While we have no reason to believe that this constitutes a random sample

of agencies, it is the full set of agencies that are available to be studied given the research design set out in the analyses we replicate. Thus, the final sample for our replication datasets includes about 9,000 agencies that report crime data and that could request SME through the federal 1033 Program.

In regards to the time period of study, the BG county-level data span from 2006 to 2012, and the HPBM county-level data span from 2000 to 2013. However, in our main results, we use replication agency-level data from 2010 to 2015 and replication county-level data from 2010 to 2014. We use these different time periods for our replication data because they represent the most recent years of data available through the more recent LESO inventory release and because of data limitations with the UCR series. Therefore our sample size is determined by the number of agencies reporting crime rates over this time period. Nonetheless, in Section 2.7 of the Supplementary Information, we also run our analyses subsetting all data to 2010 to 2012, as these are the years for which we have overlap across all of the datasets. With this smaller but more comparable set of observations, we find similar results to our main results. We still do not find that military aid reduces crime rates at either the county or agency level.

Discussion of mechanisms and limitations. One potential alternative mechanism is that an increase in equipment from the 1033 Program displaces, but does not decrease, violent crime. If this is the effect that BG and HPBM are finding, it would have to be that criminal activity is being displaced across county lines in order to see the relationship between an increase in 1033 equipment and a decrease in violent crime. (One study suggests that SME might contribute to crime displacement[20].) Additionally, this relationship should still be visible in our agency-level approach, but is not. Displacement is not the same thing as a reduction in crime, and so even if these studies' conclusions are sound, they do not support the public policy justification for the 1033 Program.

HPBM argue that their findings are the result of a deterrence mechanism whereby increasing 1033 equipment deters crime. In support of this mechanism, they discuss how the relationship between the instrument and total aid is stronger for counties that received aid for multiple years. However, this would suggest that we should see a very strong deterrence effect in our data which occurs later as agencies and counties have had 1033 equipment for potentially many more years. The fact that we find no such effect suggests either that this mechanism is not supported or that 1033 equipment can deter crime for only a few years. Even if the latter is true, it again is weak justification for the 1033 Program.

While we do not find evidence supporting the claim that an increase in 1033 equipment by police is associated with a decrease in crime, we caution against drawing strong conclusions for no association as well. With the problems and inconsistencies we have documented in the data, we would be hesitant to draw strong conclusions from any analyses using these data.

Replication of BG. To disentangle the effect of police militarization on crime from a likely endogenous relationship, BG employ an instrument that combines information about military spending by the federal government and previous acquisitions of SME by counties. Specifically, they construct a variable by interacting a county's likelihood of receiving any SME with US military spending, as a county is only eligible to receive that aid if federal military spending is sufficiently high in the previous year. They sum the quantity of items received by each county and divide that number by the lagged amount of military spending by the US government. Essentially, this allows the authors to compare counties that frequently receive aid with counties that rarely do, in years following high US military spending relative to those following lower military spending.

Following BG, we estimate the following specification:

$$\text{Crime}_{a,t} = \beta\,\hat{\text{Equipment}}_{a,t-1} + \alpha_a + \delta_{t,s} + \gamma X_{a,t} + \epsilon_{a,t} \qquad (1)$$

where $\hat{\text{Equipment}}_{a,t-1}$ is the fitted monetary value of equipment at agency a at lagged year $t-1$, from a first-stage regression. The model also includes agency fixed effects α and interactive state–time fixed effects δ. Finally, we include a matrix of demographic control variables, $X_{a,t}$, following BG: percent in poverty, median household income, unemployment, population, population male, population Black, population aged 15–19 yrs, population aged 20–24 yrs and population aged 25–34 yrs. All control variables were collected from the US Census Bureau. First-stage and reduced-form results are included in Section 2.2 of the Supplementary Information.

Next, we estimate the instrumental variables (IV) regression model using both the replication data from BG and our disaggregated information at the agency level. The summary statistics shown in Supplementary Tables 20 and 21 reveal that our county replication data do not differ very substantially from the replication data from BG, except for the militarization variables, which are plagued by the issues discussed in this article. We use the same estimation strategy for both datasets, including interactive state and year and either county or agency fixed effects. Finally, we cluster our standard errors by state, as BG do in their analysis. The results of both of these regressions are presented in Supplementary Table 1. We removed from the agency-level database outliers that had all crime rates over one million, which heavily skewed our regression results (one observation in the BG models). Results including all observations are also included in Section 2.8 of the Supplementary Information. In addition, results for controlled items only are included in Section 2.6 of the Supplementary Information.

Using the code and data provided by BG, we are able to replicate all tables in their paper in both Stata and R. All the coefficient estimates are identical, but the standard errors are slightly different between R and Stata due to the different methods used to compute clustered standard errors. However, with our replication county-level and agency-level datasets, we do not find similar negative impacts on crime rates, and instead find insignificant effects, as shown in Fig. 3.

Replication of HPBM. Like BG, HPBM recognize the potential for endogeneity in the estimation of the effect of military aid on crime rates and employ an IV approach. However, they use a different instrument. In fact, they propose four different IVs, all of which focus on the likelihood a county will receive aid based on its location. Though the federal government distributes SME to subnational agencies free of charge, agencies must pay the costs of shipping the equipment from a field activity centre (FAC), of which 18 exist around the country. Therefore, an agency's proximity to a FAC directly contributes to the likelihood that an agency will request aid, on the logic that it is cheaper for an agency closer to a FAC to get any military equipment than an agency further from it. HPBM use two IVs that measure distances to/from FACs. However, as with BG, because their data are only available aggregated by county, those instruments are measured at the the county level, rather than at the level of an individual agency that makes the decision about whether to acquire the equipment. The first instrument is the inverse distance from the closest FAC, and the second is the inverse distance from the sixth-largest FAC. The authors use two separate measures of the distance from a FAC because an item could originate from any of the 18 FACs across the country, and no one centre has all categories or types of equipment. The last two instruments consider factors that contribute to the likelihood of the agencies in a county requesting equipment, namely the county's geographic size and its degree of drug trafficking. HPBM use a county's land area and whether it was designated a High Intensity Drug Trafficking Area as additional instruments, with both measures positively influencing the probability that agencies in a county will request SME. In addition, as distinct from BG, they examine military aid in various categories of equipment, separately.

Importantly, our data differ slightly from HPBM because of data limitations at the city level. First, while they include a set of controls concerning arrest rates at the county level, not all of those data are made available at the city level. We only include that information that is listed by the FBI, so our arrest data only consider arrests made on charges of murder, rape, robbery, burglary and assault. Second, while the county data record gun assault crimes, the city data do not. So, the dependent crime variables we look at are rates of murder, robbery, assault and vehicle theft (columns 1, 2, 4 and 5 in Table 8 of HPBM). Finally, there remain notable discrepancies between the instruments used in HPBM and the ones we calculate from their description of those instruments in the paper (see Section 1.6.1 of the Supplementary Information).

Following HPBM, we estimate the following specification:

$$\text{Crime}_{a,t} = \beta_1 \hat{E}_{a,t-1} + \beta_2 X_{a,t} + \beta_3 L_{a,t-1} + \alpha_a + \nu_t + \epsilon_{a,t} \qquad (2)$$

The outcome of interest in the second-stage regression is the reported crime rates—for the crimes of homicide, robbery, assault and vehicle theft—for agency a at time t. Note that the HPBM analysis also examines crime rates for gun assaults. However, the agency-level data do not contain information on this variable, so it is excluded from the results reported here. The variable $\hat{E}_{a,t-1}$ is the estimate of the lagged equipment for jurisdiction a at $t-1$. In this analysis, we examine two variables regarding equipment: either the logged number of items shipped in the previous year, or the logged value of those shipments. Second, X is a matrix of controls including per capita personal income, unemployment and population (all collected from the US Census Bureau), and $L_{a,t-1}$, the lagged arrest rates for the crimes of murder, rape, robbery, burglary and assault (from UCR). Finally, α and ν represent vectors of agency and year fixed effects, respectively. The first-stage and reduced-form results are included in Section 2.4 of the Supplementary Information.

Next, we estimate the IV regression model using both the replication data from HPBM and our disaggregated information at the agency level. Similar to the BG replication, the summary statistics shown in Supplementary Tables 20 and 21 reveal that our county replication data do not differ notably from the replication data from HPBM, except for the militarization variables, which again suffer from the issues discussed in this article. We use the same estimation strategy for both datasets, including additive year and either county or agency fixed effects. Finally, we cluster our standard errors by either county or agency. The results of both of these regressions are presented in Supplementary Table 2. As with the BG replication, we removed from the agency-level database outliers that had all crime rates over one million, which heavily skewed our regression results (three observations in the HPBM models). Results including all observations are included in Section 2.8 of the Supplementary Information. In addition, results for controlled items only are included in Section 2.6 of the Supplementary Information.

Using the code and data provided by HPBM, we are able to replicate all tables in their paper in both Stata and R. All the coefficient estimates are identical, but the standard errors are slightly different between R and Stata due to the different

methods used to compute the clustered standard errors. However, as with the replication of BG, we do not find similar negative impacts on crime rates using our replication county-level and agency-level datasets. Instead, we find either insignificant effects or large and significant positive impacts on crime rates, as shown in Fig. 4.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are publicly available through our Dataverse, found at https://dataverse.harvard.edu/dataverse/politicsofpolicinglabnaturemilitarization/. Data are publicly available, but users may be required to create an account.

Code availability

The code that support the findings of this study is publicly available through our Dataverse, found at https://dataverse.harvard.edu/dataverse/politicsofpolicinglabnaturemilitarization/. The code is publicly available, but users may be required to create an account.

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Author contributions

A.G. contributed to the research design, analysis and collection of data, and is the lead author. E.C. and K.J.S. contributed to the analysis and collection of data and writing. T.S.C. contributed to development of the research idea, research design, collection of data, interpretation of the analysis, and writing. A.N.G. contributed to development of the research idea, research design, direction of the analysis, interpretation of the analysis, and writing. M.L.O. contributed to development of the research idea, the literature review and writing.

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to T.S.C.

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Research sample	We primarily use two existing datasets: the 1033 Program data publicly available through the DLA. We use the version from March 31, 2018. Link to the current version of the data: https://www.dla.mil/DispositionServices/Offers/Reutilization/LawEnforcement/PublicInformation/ Second, we use the FBI Uniform Crime Reports data series for city and county crime rates. Link to the website: https://www.fbi.gov/services/cjis/ucr
Sampling strategy	Our sample size is determined by the number of agencies reporting crime rates over this time period. We then link those agencies that report crime rates to DLA militarization records.
Data collection	We collected the data from the public sources identified above and stored them on a shared Dropbox.
Timing	We collected the data from the public sources identified above in spring 2018 onto a shared Dropbox.
Data exclusions	In our main analyses, we exclude agencies with crime rates over 1 million (1-3 observations), though we include all observations in our supplementary materials. Otherwise, no data is excluded from the two public datasets identified above (pending that they matched accurately across datasets).
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